

Task 2.3 – Medial Axis Detection Documentation

Abstract

This project detects the medial axis of a moving surgical tool from video frames. A computer vision pipeline consisting of background subtraction, morphological processing, edge detection, and a custom Hough Transform was implemented.

Introduction

The medial axis represents the central skeletal line of an object and is useful for analyzing object orientation and structure. The task involves processing video frames to detect this axis for a moving surgical tool.

Problem Statement

Detect the medial axis of a moving surgical tool using computer vision techniques. A custom implementation of the Hough Transform must be used instead of built-in OpenCV functions.

Processing Pipeline

- Frame extraction from the video.
- Background subtraction to isolate the moving object.
- Morphological cleaning to remove noise.
- Edge detection using Sobel derivatives.
- Custom Hough Transform to detect straight edges.
- Compute medial axis between detected edges.

Results

The algorithm successfully detects the edges of the tool and overlays the medial axis on the original frames, enabling structural analysis of the object.

Conclusion

The proposed pipeline effectively identifies the medial axis by combining background subtraction, morphological processing, edge detection, and a custom Hough Transform.